

COURSE TITLE : PROGRAMMING IN JAVA
COURSE CODE : 3262
COURSE CATEGORY : B
PERIODS/WEEK : 5
PERIODS/SEMESTER : 75
CREDITS : 5

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Programming Overview of Java	21
2	Inheritance, Packages, Exception Handling	18
3	Creating GUI with AWT and Swing	18
4	Applets & Application	18

COURSE GENERAL OUTCOMES:

Sl.	G.O	On completion of this course the student will be able to :
1	1	To understand the concepts of data type, variable and control structures in JAVA
	2	To understand the concept of classes, objects and methods
2	1	To demonstrate inheritance, Packages and Interfaces.
	2	To appreciate Exception handling in JAVA.
3	1	To construct GUI using AWT and SWING
4	1	To use Java Applets and inter applet communication

SPECIFIC OUTCOMES:

MODULE – I: Programming Overview of Java

1.1 To understand the concepts of data type, variable and control structures in JAVA

1.1.1 To state the basic concept of JAVA

- 1.1.2 To illustrate the features of JAVA
- 1.1.3 To Identify the data types and variables in JAVA
- 1.1.4 To describe the operators and different Control Structures in JAVA
- 1.1.5 To discuss arrays and array handling in JAVA
- 1.1.6 To develop JAVA programs using the basic programming concepts of JAVA
- 1.2 To understand the concept of classes, objects and methods
 - 1.2.1 To illustrate the creation and implementation of classes in Java
 - 1.2.2 To describe the concept of Static Classes in JAVA

MODULE – II: Inheritance, Packages, Exception Handling

- 2.1 To demonstrate Inheritance, Packages and Interfaces.
 - 2.1.1 To describe the concepts of Base class and derived class
 - 2.1.2 To identify the different types of Inheritance in JAVA
 - 2.1.3 To discuss final methods
 - 2.1.4 To discuss and implement inner classes and anonymous inner class in JAVA
 - 2.1.5 To describe the concept of interfaces in JAVA
 - 2.1.6 To demonstrate the concept of packages in JAVA
- 2.2 To appreciate Exception handling in JAVA.
 - 2.2.1 To discuss the Exception Handling in JAVA
 - 2.2.2 To demonstrate Exception Handling in JAVA

MODULE – III: Creating GUI with AWT and Swing

- 3.1 To construct GUI using AWT and SWING.
 - 3.1.1 To describe about AWT and components
 - 3.1.2 To describe SWING and its components.
 - 3.1.3 To use the AWT components, Containers and layouts
 - 3.1.4 To use the SWING components, Containers and layouts
 - 3.1.5 To describe the JDK1.1 event model
 - 3.1.6 To apply the event listeners and event handling mechanisms
 - 3.1.7 To define and use adapter classes.

MODULE – IV: Applets & Application

- 4.1 To use Applets and inter applet communication
 - 4.1.1 To discuss applets and applet life cycle
 - 4.1.2 To construct simple applets

- 4.1.3 To recognize the various applet tags
- 4.1.4 To distinguish Applets & Application
- 4.1.5 To illustrate the security features of Applets
- 4.1.6 To use Java programming language features to design and create Java applets.
- 4.1.7 To describe Inter applet communication.

CONTENT DETAILS

MODULE – I: Programming Overview of Java

An overview of Java, Features of Java, data types, variables and arrays, operators, control statements, Creating & using classes in Java – Static classes

MODULE – II: Inheritance, Exception Handling

Inheritance, Final methods, Inner Classes, Anonymous inner classes, Packages and Interfaces, Exception handling, Creating and using exceptions

MODULE – III: Creating GUI with AWT and Swing

Creating GUI with AWT and Swing – AWT- components, containers, layout, JDK, eventmodel, event handling, Event listeners, Adapter classes.

MODULE – IV: Applets & Application

Applets & Application – Life cycle of applets - Security features for applets - Inter applet communication, development and execution, applet tag

TEXT BOOK(S)

1. Herbert Schildt, “The Java 2 : Complete Reference”, Fourth edition, TMH, 2002

REFERENCE

1. Rajkumar Buyya, Selvi, Chu. “Object oriented programming with JAVA essentials and applications” Mc Graw Hill
2. H.M. Deitel, P.J. Deitel, "Java : how to program", Fifth edition, Prentice Hall of India private limited.
3. The Java Programming Language 3rd Edition - Arnold, Gosling, Holmes, Pearson Education Asia
4. Using Java 2 Platform - Joseph Weber, PHI